EXP NO:17 Experiment to understand Linear Regression for a given data set.

AIM:
To understand the Linear Regression for the data set given.

PROGRAM:

python

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Creating dataset: hours studied vs marks
data = {
    'Hours': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Marks': [35, 40, 48, 52, 58, 63, 70, 75, 82, 88]
}
df = pd.DataFrame(data)
df

# Information about the dataset
df.info()

# Scatter plot of the data
plt.scatter(df['Hours'], df['Marks'])
plt.xlabel("Hours Studied")
plt.ylabel("Marks")
plt.title("Hours Studied vs Marks")
plt.show()

# Select feature and label
X = df[['Hours']]
y = df['Marks']

# Split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create and train the Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Slope and intercept
print("Slope (Coefficient):", model.coef_[0])
print("Intercept:", model.intercept_)

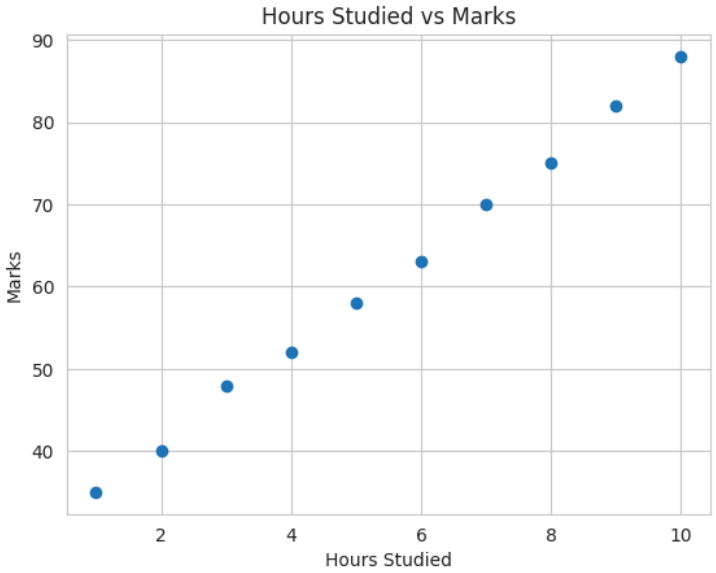
# Predict on test data
y_pred = model.predict(X_test)
print("Predicted values:", y_pred)
print("Actual values:", y_test.values)

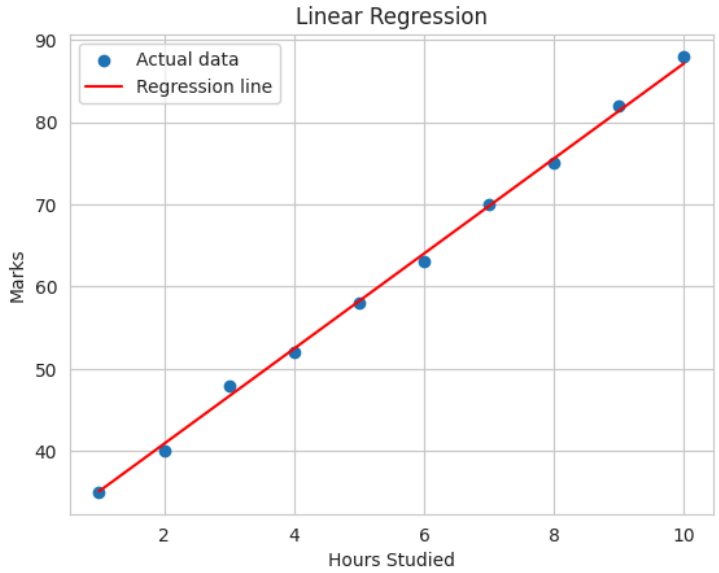
# Evaluate the model
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

# Plot the regression line
plt.scatter(df['Hours'], df['Marks'], label="Actual data")
plt.plot(df['Hours'], model.predict(X), color='red', label="Regression line")
plt.xlabel("Hours Studied")
plt.ylabel("Marks")
plt.title("Linear Regression")
plt.legend()
plt.show()

# Predict marks for a new student who studied 7.5 hours
new_hours = pd.DataFrame({'Hours': [7.5]})
print("Predicted Marks for 7.5 hours:", model.predict(new_hours)[0])
```

OUTPUT:





RESULT:

Thus the program to understand Linear Regression for the given data set was executed successfully.